Rate of reaction: Rate of reaction can be measured in terms of either decrease in concentration of any one of the reactants or increase in concentration of any one of the products per unit time.

Factors which affect the rate of the reaction are concentration, temperature and catalyst.

Sodium thiosulphate reacts with hydrochloric acid and produces a colloidal solution of sulphur, which makes the solution translucent. The reaction occurs as follows :

$$Na_2S_2O_3(aq) + 2HCl(aq) \longrightarrow 2NaCl(aq) + H_2O(l) + SO_2(g) + S(s)\downarrow$$

Ionic form of the above reaction is written as :

$$S_2O_3^{2-}(aq) + 2H^+(aq) \longrightarrow H_2O(l) + SO_2(g) + S(s)\downarrow$$

The property of the colloidal solution of sulphur to make the solution translucent is used to study the rate of precipitation of sulphur. The rate of precipitation of sulphur increases with increase in the concentration of the reacting species or with increase in the temperature of the system. With increase in the concentration of reactants, the number of molecular collisions per unit time between the reacting species increases and consequently, the chances of product formation increase. This result increase in the rate of precipitation of sulphur. Similarly, on increasing the temperature, the kinetic energy of the reacting species increases. So the number of effective collisions that result in the formation of products increase leading to a faster rate of reaction.

Experiment No. 3

Date: / /

Aim: To study the effect of concentration on the rate of reaction between sodium thiosulphate and hydrochloric acid.

Theory: Law of mass action states that rate of a chemical reaction is directly proportional to the product of active masses (molar concentrations) of reactants. The rate of reaction between sodium thiosulphate and hydrochloric acid depends on the concentration of reactants. As the concentration of reactants increses, the rate of reaction also increases. The balanced chemical reaction is,

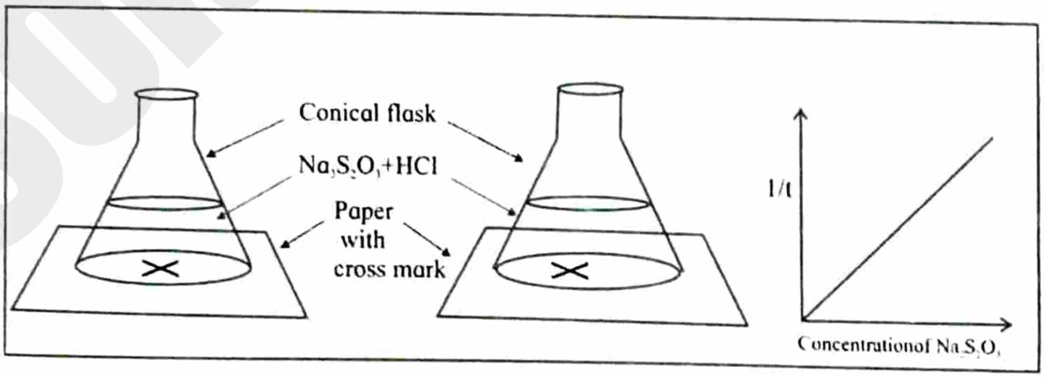
$$Na_2S_2O_3(aq) + 2HCl(aq) \longrightarrow S(s)\downarrow + NaCl(aq) + SO_2(g) + H_2O(l)$$

Sulphur formed during the reaction is insoluble and forms milky non transparent opaque colloidal solution. The time required to produce enough sulphur, so that cross mark on the paper kept below the conical flask can not be seen, when observed from top and hence rate of reaction be calculated.

Apparatus: Burette (50 mL), conical flask (250 mL), stop watch, plane paper with cross mark, measuring cylinder (10 mL), etc.

Chemicals: 0.1 M $Na_2S_2O_3$, 1 M HCl, distilled water.

Diagram:



Procedure:

1. Take four conical flasks (250 mL), wash with water and label them as A, B, C, D respectively.
2. Using 50 mL burette take exactly 20, 30, 40 and 50 mL 0.1 M $Na_2S_2O_3$ in the flasks A, B, C, D respectively.
3. Using another 50 mL burette add 28, 18 and 8 distilled water to flasks A, B and C respectively. (There is no addition of distilled water in flask D)
4. With the help of 10 mL measuring cylinder, add 2 mL 1 M HCl to the flask A containing 20 mL 0.1 M $Na_2S_2O_3$ and 28 mL distilled water. Start the stop watch immediately.
5. Shake and keep the conical flask on a paper having cross mark and view the cross mark through reaction mixture from top of the conical flask.
6. When the cross mark on the paper just become invisible stop the stop watch immediately and record the time required in seconds.
7. Repeat the experiment by adding 2 mL 1 M HCl to flasks B, C and D, simultaneously. By using stop watch record the time required, when cross mark on the paper just becomes invisible. (Use same paper having cross mark for flasks B, C and D)

Observation Table:

Conical Flasks	0.1 M $Na_2S_2O_3$ in mL	H_2O mL	1M HCl mL	Time required to become cross mark invisible (t) sec	1/t s^{-1}	Concentration of $Na_2S_2O_3$
A	20	28	2	125	0.008	0.04
B	30	18	2	100	0.010	0.06
C	40	8	2	83	0.012	0.08
D	50	-	2	71	0.014	0.1

Graph: Plot a graph of 1/t against concentration of sodium thiosulphate solution.

Result:

Rate of reaction (1/t) is Between $Na_2S_2O_8$ & HCl is directly Proportional to the concentration of $Na_2S_2O_3$ Because 1/t is direct measure rate of reaction Rate of reaction $\propto [Na_2S_2O_3]$

MCQ

Select the correct answer for the following multiple choice type questions.

1. The rate of reaction between 0.1 M $Na_2S_2O_3$ and 1 M HCl does NOT depends on
   a. temperature b. concentration
   c. pressure d. catalyst
2. Sodium thiosulphate solution reacts with hydrochloric acid solution to produce ....
   a. a colloidal solution of sulphur b. transparent solution of sulphur
   c. black solution of sulphur d. white precipitate of NaCl
3. The time required for completion of the reaction, when we add 2 mL of 1M HCl to the flasks A, B, C and D containing increasing order of concentration of 0.1 M $Na_2S_2O_3$ ....
   a. increases b. decreases
   c. first decreases and then increases d. first increases and then decreases
4. The nature of graph for effect of concentration on rate of reaction between 0.1 M $Na_2S_2O_3$ and 1 M HCl is a straight line with ....
   a. decreasing slope b. increasing slope
   c. increasing slope intersecting to y-axis d. decreasing slope intersecting to x-axis

Short answer questions

1. Define rate of reaction.

Ans. Rate of reaction is defined as the speed of which reactants are converted into produce or the spend with which a reaction take places

2. Mention the factors affecting the rate of reaction.

Ans. Concentration of Reactent, nature of reaclent, surface area of reatent, catagst, temperature pressure these factors affects the rate of reaction.

3. Write the name of law for the study of effect of concentration on the rate of reaction.

Ans. A differential rate law is used for the study of effect of concentration on the rate of reaction.

4. Explain the nature of graph of 1/t against concentration of $Na_2S_2O_3$ solution.

Ans. The nature of graph of 1/t against concentration of Na2S2O3 solution is a (striight line graph)

5. Why rate of reaction is fastest, if distilled water is not added to 50 mL (i.e D flask) 0.1 M $Na_2S_2O_3$?

Ans. The concentration of the reactent plays an important role in the rate of reaction As the condentration of reacent the no. of nacting molecule increase B'cause the no. of corisions also (bemo as reaction rate increase)

Remark and sign of teacher: ....